

Patient Management Portal - Non-Functional Requirements v1

Confidential - Internal Use Only

Version 1.0

1. Performance Requirements

NFR-HCP-001: Patient dashboard page load time < 2 seconds at P95 on broadband connections.

NFR-HCP-002: Appointment calendar availability query must return results within 800ms for a 30-day window.

NFR-HCP-003: Lab results page must render within 1.5 seconds even with 200+ historical results.

NFR-HCP-004: Video consultation connection establishment < 3 seconds from click-to-first-frame.

2. Scalability & Capacity

NFR-HCP-005: System must support 5,000 concurrent patient sessions during peak hours (Monday 8-10am).

NFR-HCP-006: Telehealth service must handle 200 simultaneous video sessions without quality degradation.

NFR-HCP-007: Notification service must process 10,000 SMS/email per hour for appointment reminders.

NFR-HCP-008: Database must handle 500 concurrent write operations during morning check-in surge.

3. HIPAA Compliance & Security

NFR-HCP-009: Annual HIPAA compliance audit must pass with zero high-risk findings.

NFR-HCP-010: All API endpoints must enforce HTTPS (TLS 1.2+). HTTP connections must be rejected, not redirected.

NFR-HCP-011: Automatic session logout after 15 minutes of inactivity with 2-minute warning dialog.

NFR-HCP-012: Role-based access control (RBAC) with 4 roles: Patient, Provider, Staff, Admin. Minimum privilege principle enforced.

NFR-HCP-013: PHI data must never appear in application logs. Log sanitization middleware required on all services.

4. Accessibility Requirements

NFR-HCP-014: Portal must comply with WCAG 2.1 Level AA. All images must have alt text. Forms must have associated labels.

NFR-HCP-015: Screen reader compatibility tested with NVDA (Windows) and VoiceOver (macOS/iOS).

NFR-HCP-016: Color contrast ratio minimum 4.5:1 for normal text, 3:1 for large text. All status indicators must not rely solely on color.

NFR-HCP-017: All interactive elements reachable via keyboard. Visible focus indicator on all focusable elements.

5. Availability & Disaster Recovery

NFR-HCP-018: System uptime SLA: 99.95% (max 21.9 minutes downtime per month). Planned maintenance windows: Sunday 2-4am ET.

NFR-HCP-019: Multi-AZ Azure deployment with automatic failover. RTO < 15 minutes, RPO < 5 minutes.

NFR-HCP-020: Telehealth service must have independent failover to backup Zoom region to prevent single point of failure.

6. Interoperability

NFR-HCP-021: HL7 FHIR R4 compliance validated using Touchstone test suite before each major release.

NFR-HCP-022: Surescripts certification renewal annually. Must pass all 36 mandatory validation scenarios.

NFR-HCP-023: X12 EDI transactions must comply with HIPAA 5010 standards. Validation via Edifecs before submission.